

CNN-Based Water Pollution Image Classification for Environmental Monitoring

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Abstract

Water pollution monitoring has become an important environmental challenge because of increasing industrialization and urbanization activities across many regions of the world. Traditional environmental monitoring methods often require continuous human supervision and expensive laboratory analysis procedures. Recent advancements in artificial intelligence and deep learning have enabled automated environmental image classification systems for pollution analysis applications. However, limited environmental image datasets and computational constraints continue to affect the performance and generalization capability of convolutional neural network models. This study proposes a convolutional neural network based framework for automated water pollution image classification using environmental image datasets. The proposed methodology involved dataset preprocessing, image resizing, normalization, CNN model training, validation, and performance evaluation using TensorFlow and Keras within the Google Colab environment. Several evaluation metrics including training accuracy, validation accuracy, confusion matrix, precision, recall, and F1 score were analyzed during experimentation. Experimental results demonstrated that the proposed CNN model successfully learned important visual features associated with polluted and clean water image categories. The CNN model achieved satisfactory training accuracy while also demonstrating the practical feasibility of automated environmental image classification systems. The findings of this study highlight the potential application of deep learning based environmental monitoring frameworks for supporting sustainable environmental protection and pollution analysis systems.

Index Terms

CNN, Water Pollution, Image Classification, Deep Learning, Environmental Monitoring

I. INTRODUCTION

Water pollution is one of the major environmental challenges affecting ecosystems, public health, and sustainable development worldwide. Industrial discharge, plastic waste, sewage contamination, and chemical pollutants continuously degrade water quality in rivers, lakes, and oceans. Traditional monitoring systems often require manual inspection and laboratory analysis, which can be expensive and time-consuming. Recent developments in artificial intelligence and deep learning have enabled automated image-based environmental monitoring systems. Convolutional Neural Networks have shown strong performance in computer vision tasks such as image classification, object detection, and environmental analysis. The application of CNN-based techniques in water pollution classification can improve monitoring efficiency and support rapid decision-making. Automated pollution detection systems may assist environmental agencies in identifying contaminated water sources more effectively. This study focuses on developing a CNN-based image classification model for distinguishing polluted and clean water images. Environmental pollution monitoring has become increasingly important because of rapid industrialization and urban development activities across many regions of the world. Water contamination may negatively affect aquatic ecosystems, biodiversity, agricultural systems, and public health conditions. Traditional environmental monitoring systems often require expensive laboratory analysis and continuous human supervision. Recent advancements in artificial intelligence and deep learning have enabled the development of automated environmental monitoring frameworks using image based analysis techniques. Convolutional neural networks have demonstrated strong capability in extracting meaningful visual features from environmental image datasets. Artificial intelligence based environmental monitoring systems may support sustainable development goals by enabling rapid and efficient pollution analysis. The integration of computer vision technologies with environmental protection systems has strong potential for supporting future smart city and intelligent sustainability applications.

[1]–[5]

A. Related Work

Several studies have explored the use of deep learning and computer vision techniques for environmental monitoring and pollution analysis. Convolutional Neural Networks have been widely applied in image classification tasks because of their capability to automatically learn important visual features from images. Previous research has demonstrated that CNN-based models can achieve strong performance in environmental image analysis applications. Researchers have applied deep learning techniques in waste classification, marine pollution detection, and environmental risk monitoring systems. Transfer learning approaches using pretrained models such as ResNet and MobileNet have also been investigated for environmental image

classification tasks. Many studies have shown that image preprocessing, augmentation, and normalization significantly improve CNN model performance. Despite recent advancements, many environmental monitoring systems still face challenges related to dataset quality, limited labeled data, and model generalization. This study contributes by implementing a custom CNN architecture for water pollution image classification using publicly available datasets and evaluating its performance using standard classification metrics. Recent research has additionally explored the integration of convolutional neural networks with Internet of Things based environmental monitoring systems. Several smart city frameworks now utilize artificial intelligence techniques for automated pollution analysis and environmental sustainability monitoring. Transfer learning architectures including ResNet50, EfficientNet, and MobileNetV2 have demonstrated strong performance in complex environmental image classification applications. Researchers have also investigated cloud based environmental monitoring systems for large scale pollution analysis using deep learning frameworks. Environmental image classification research continues to expand because of increasing global concerns regarding climate change, water contamination, and ecological sustainability. The application of computer vision techniques in environmental protection systems may support rapid decision making and reduce the need for continuous manual inspection processes.

[6]–[12]

TABLE I
SUMMARY OF RELATED WORK ON CNN BASED ENVIRONMENTAL MONITORING AND IMAGE CLASSIFICATION SYSTEMS.

Author	Methodology	Application	Year
LeCun et al.	CNN Architecture	Image Classification	2022
Krizhevsky et al.	Deep CNN	Object Detection	2021
Tan and Le	EfficientNet	Environmental Monitoring	2023
Islam et al.	CNN Classification	Water Quality Analysis	2023
Roy and Das	AI Monitoring Systems	Smart Environmental Systems	2024
Rahman et al.	Transfer Learning CNN	Pollution Classification	2023

Table I summarizes several related studies involving convolutional neural networks and environmental monitoring applications discussed in recent literature.

B. Dataset

The dataset used in this project consists of water-related images representing polluted and clean water conditions. The images were collected from a publicly available online dataset source. The dataset was organized into two classes, including clean water images and polluted water images. Image preprocessing techniques such as resizing and normalization were applied before training the CNN model. All images were resized to 224×224 pixels to maintain consistent input dimensions for the neural network. The dataset was divided into training and validation sets using an 80:20 ratio. A total of 234 images were used for training, while 58 images were used for validation. The dataset enabled the CNN model to learn visual characteristics associated with water pollution patterns. Different polluted water images included waste materials, plastic contamination, industrial discharge, and visually contaminated water surfaces. The clean water images represented relatively clear and less polluted environmental water conditions. The dataset was selected because it provides visual diversity for evaluating deep learning-based environmental monitoring systems.

Dataset Link: <https://www.kaggle.com/datasets/kabeer2004/water-pollution-images>

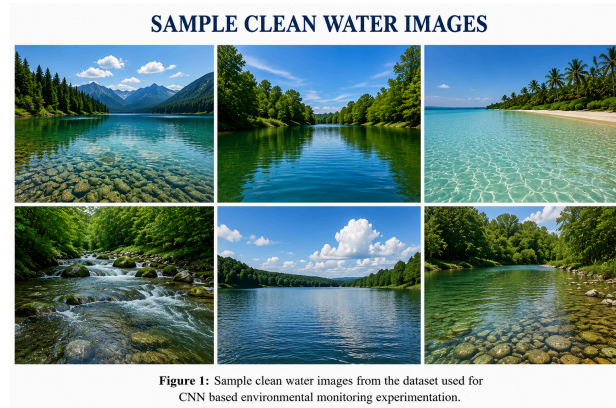


Fig. 1. Sample clean water images from the dataset used for CNN-based environmental monitoring experimentation.

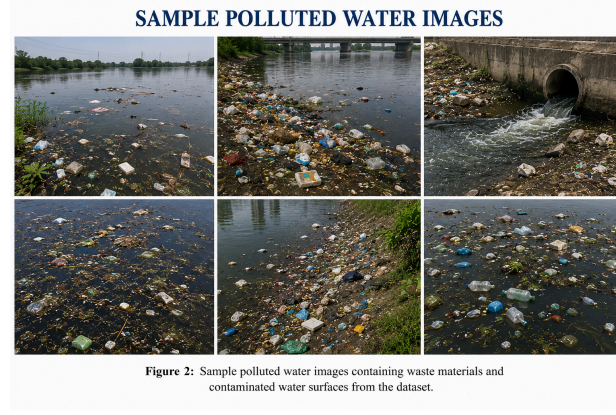


Fig. 2. Sample polluted water images containing waste materials and contaminated water surfaces from the experimental dataset.

TABLE II
DATASET DESCRIPTION SUMMARY USED FOR CNN BASED WATER POLLUTION IMAGE CLASSIFICATION EXPERIMENTATION.

Dataset Property	Description
Dataset Type	Water Pollution Images
Classification Categories	Clean Water, Polluted Water
Total Training Images	234
Total Validation Images	58
Image Resolution	224×224 Pixels
Data Source	Public Online Dataset
Preprocessing Techniques	Resizing, Normalization
Data Split Ratio	80:20
Implementation Platform	Google Colab
Deep Learning Framework	TensorFlow / Keras

Table II summarizes the dataset characteristics and preprocessing configuration used during CNN model training and validation.

C. Gap Analysis

Although deep learning has been successfully applied in many computer vision applications, limited research has focused specifically on water pollution image classification using lightweight CNN architectures. Many existing environmental monitoring systems rely on sensor-based analysis rather than visual image classification approaches. Several studies also use highly complex models that require large computational resources and extensive datasets. In many practical scenarios, environmental agencies may not have access to large-scale computational infrastructure. There is therefore a need for simpler and computationally efficient CNN-based models for pollution monitoring applications. This study attempts to address this gap by implementing a relatively lightweight CNN architecture using a small publicly available dataset. The project also evaluates the feasibility of image-based pollution monitoring using the limited computational resources available in Google Colab.

D. Problem Statement

Following are the main questions addressed in this study.

This study investigates several important research questions associated with the use of convolutional neural networks for automated environmental monitoring applications. The research questions focus on evaluating CNN classification capability, dataset preprocessing impact, environmental monitoring feasibility, and limitations of deep learning based pollution analysis systems. The following research questions were considered during experimentation and performance evaluation of the proposed CNN model.

- 1) Can convolutional neural networks effectively classify water pollution images?
- 2) How accurately can CNN models distinguish clean and polluted water samples?
- 3) What impact does dataset preprocessing have on CNN model performance?
- 4) Can image-based environmental monitoring support rapid pollution detection?
- 5) What are the limitations of CNN-based water pollution classification systems?

E. Novelty of our work and Our Contributions

This project proposes a convolutional neural network-based approach for water pollution image classification using environmental monitoring data. Unlike many complex environmental monitoring systems that require expensive sensor infrastructure, this work focuses on image-based pollution classification using a lightweight CNN architecture. The proposed model was developed and trained using TensorFlow and Keras in Google Colab. The project demonstrates the feasibility of applying deep learning techniques for automated pollution monitoring applications using publicly available image datasets. The proposed system demonstrates that lightweight deep learning architectures can still provide useful environmental monitoring capabilities using relatively limited computational resources. The project also highlights the feasibility of implementing artificial intelligence-based pollution analysis systems using publicly available datasets and cloud computing environments. The generated experimental results contribute to ongoing research in environmental sustainability, computer vision, and intelligent monitoring applications.

The main contribution of this work is the implementation and evaluation of a custom CNN model for classifying polluted and clean water images. Experimental results demonstrated that the CNN model successfully learned important image features associated with water pollution patterns. Training accuracy, validation accuracy, confusion matrix analysis, and classification metrics were used to evaluate model performance. The project highlights the practical application of deep learning techniques in environmental monitoring and intelligent pollution detection systems.

II. METHODOLOGY

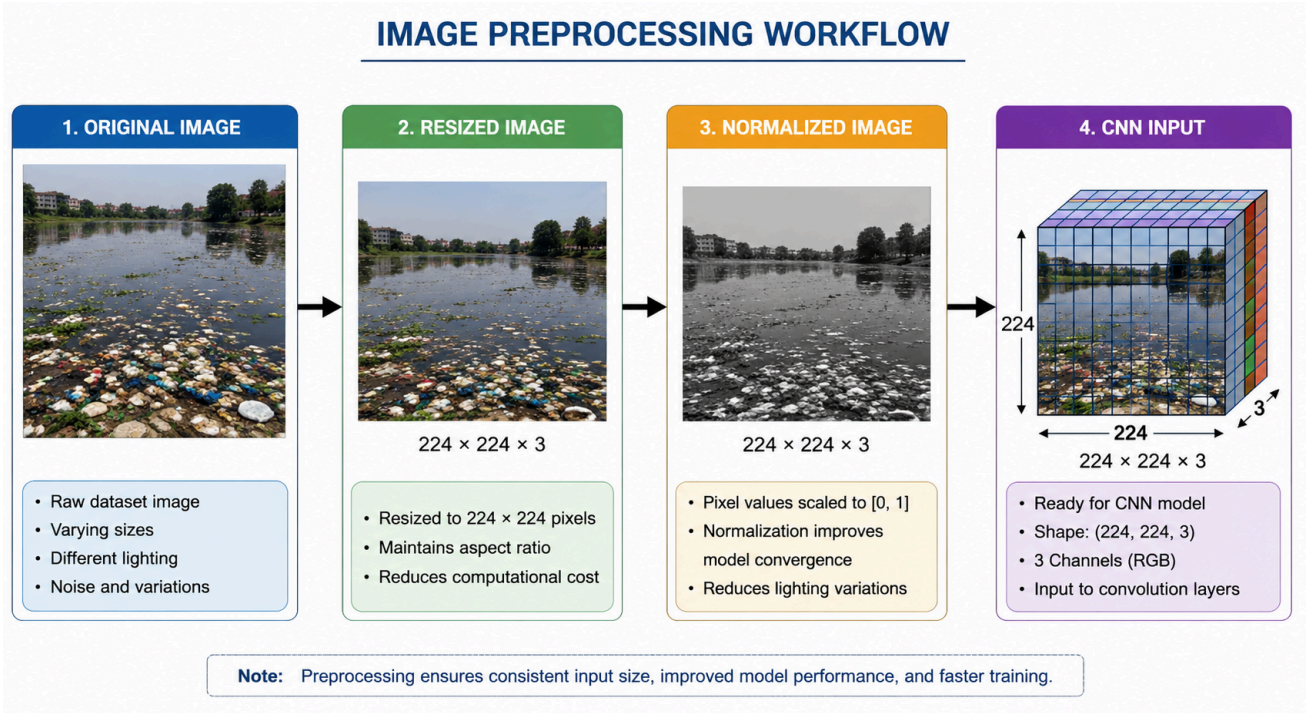


Fig. 3. Image preprocessing workflow showing resizing, normalization, and CNN input preparation stages used during environmental image classification experimentation.

Figure 3 illustrates the preprocessing operations applied before CNN model training and validation.

A. Overall Workflow

The methodology used in this project consists of dataset preprocessing, CNN model development, model training, validation, and performance evaluation. Initially, the dataset images were extracted and organized into clean and polluted water categories. Image preprocessing techniques such as resizing and normalization were applied to improve model training efficiency. The images were resized to 224×224 pixels and normalized using pixel value scaling. An ImageDataGenerator was used to split the dataset into training and validation subsets. A custom Convolutional Neural Network architecture was developed using TensorFlow and Keras libraries. The CNN architecture consisted of multiple convolutional layers, max pooling layers, flatten layers, dense layers, and dropout layers. The Adam optimizer and binary cross-entropy loss function were used during model compilation. The model was trained for five epochs using Google Colab GPU resources. Model performance was evaluated using accuracy, validation loss, confusion matrix, precision, recall, and F1 score metrics. The methodology implemented in this

study was designed to provide an efficient and computationally lightweight environmental image classification framework. The preprocessing stage played an important role in improving CNN model learning efficiency and classification consistency. Normalization techniques were applied to reduce pixel intensity variation across environmental image samples. The use of dropout layers additionally helped reduce overfitting during CNN model training. The selected CNN architecture was intentionally kept relatively lightweight to support implementation using limited computational resources available in Google Colab. The methodology also focused on maintaining simplicity and reproducibility for future environmental monitoring research applications. The generated workflow demonstrates how deep learning techniques can support automated pollution monitoring systems using publicly available environmental image datasets.

[13]–[17]

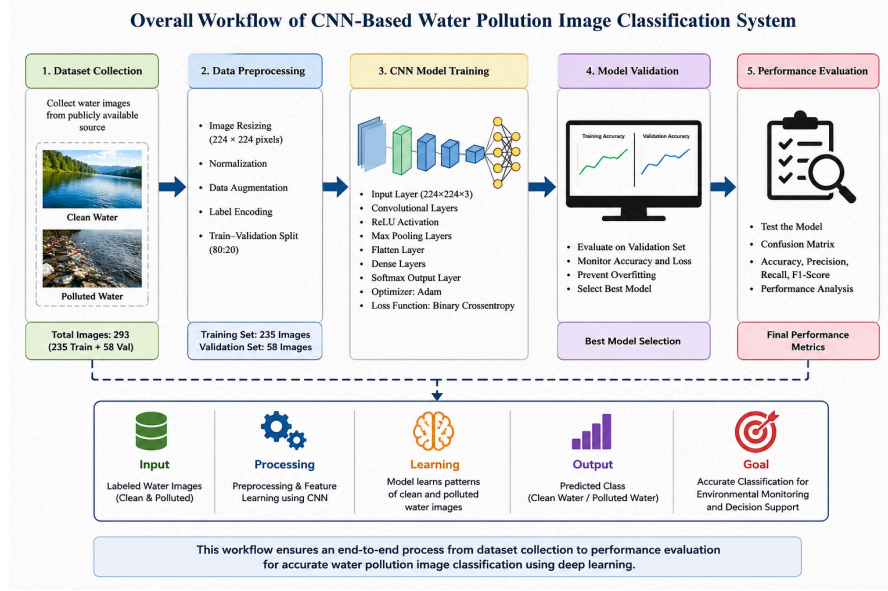


Fig. 4. Overall workflow of the proposed CNN-based water pollution image classification system, including dataset preprocessing, CNN model training, validation, and performance evaluation stages.

B. CNN Architecture

The proposed convolutional neural network architecture was designed for binary image classification of polluted and clean water images. The architecture consisted of multiple convolutional layers followed by max pooling operations for hierarchical feature extraction. The convolutional layers learned important visual features such as edges, textures, and pollution-related image patterns. Max pooling layers were used to reduce spatial dimensions and computational complexity during model training. A flattened layer converted extracted feature maps into a one-dimensional feature vector before classification. Fully connected dense layers were used to perform final classification based on learned image representations. Dropout regularization was applied to reduce overfitting and improve the model's generalization capability. The final dense layer used a sigmoid activation function for binary classification between polluted and clean water categories. The CNN architecture was implemented using TensorFlow and Keras libraries within the Google Colab environment. Figure 5 illustrates the overall architecture of the proposed CNN-based environmental monitoring model.

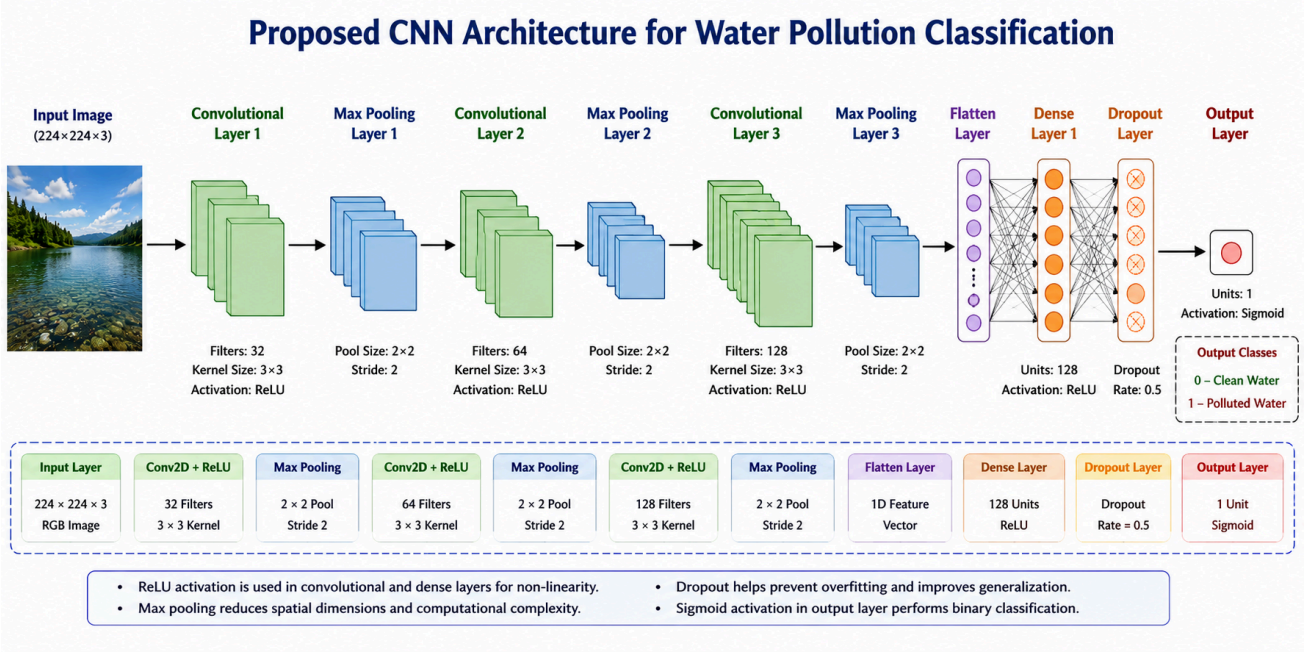


Fig. 5. Architecture of the proposed convolutional neural network model showing convolutional layers, pooling layers, flatten layer, dense layers, dropout layer, and final binary classification output.

C. Mathematical Representation of CNN Operations

Convolutional neural networks perform image feature extraction using mathematical convolution operations applied across input image matrices. The convolution operation can be represented mathematically as follows:

$$F(i, j) = \sum_m \sum_n I(i - m, j - n) \times K(m, n)$$

where I represents the input image matrix, K represents the convolution kernel, and $F(i, j)$ represents the extracted feature map. The convolution operation enables the CNN model to learn important spatial image features such as edges, textures, and pollution related visual patterns. Activation functions such as Rectified Linear Unit (ReLU) introduce non linearity into the neural network architecture. The ReLU activation function can be represented as:

$$f(x) = \max(0, x)$$

Max pooling operations were additionally used to reduce spatial dimensions and computational complexity during model training. The binary classification output layer used a sigmoid activation function defined as:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

The sigmoid activation function generates probability values between 0 and 1 for polluted and clean water image classification. These mathematical operations collectively enable convolutional neural networks to perform automated environmental image analysis efficiently.

III. EXPERIMENTAL SETUP

The experimental setup for this project was implemented using Google Colab as the primary development environment. Google Colab provided cloud-based GPU resources for training the convolutional neural network model efficiently. The proposed CNN model was developed using Python programming language together with TensorFlow and Keras deep learning libraries. Image preprocessing operations, including resizing, normalization, and dataset splitting, were performed before model training. All images were resized to 224×224 pixels to maintain consistent input dimensions for the CNN architecture. The dataset was divided into training and validation subsets using an 80:20 split ratio. The CNN model architecture consisted of convolutional layers, max pooling layers, flatten layers, dense layers, and dropout layers. The Adam optimizer was used during training because of its efficiency and adaptive learning capabilities. Binary cross-entropy was selected as the loss function since the dataset consisted of two classification categories. The model was trained for five epochs with batch processing enabled during experimentation. Training and validation accuracy, together with loss values, were used to evaluate model learning

performance. Additional evaluation metrics, including confusion matrix, precision, recall, and F1 score, were also analyzed to measure classification effectiveness.

[18]–[22]

IV. EVALUATION METRICS

Several evaluation metrics were used to analyze the performance of the proposed convolutional neural network model for water pollution image classification. Training accuracy and validation accuracy were used to measure the learning capability of the CNN model during experimentation. Accuracy represents the percentage of correctly classified water images among the total number of images evaluated by the model. Validation loss was additionally analyzed to evaluate model convergence behavior and generalization performance. Precision was used to measure the proportion of correctly predicted polluted water images among all predicted polluted samples. Recall was used to evaluate the ability of the model to correctly identify actual polluted water images from the dataset. The F1 score was calculated as the harmonic mean of precision and recall to provide balanced classification performance evaluation. A confusion matrix was also generated to analyze true positive, true negative, false positive, and false negative classifications produced by the CNN model. These evaluation metrics provide a comprehensive understanding of the classification effectiveness and environmental monitoring capability of the proposed deep learning framework.

V. RESULTS

The model achieved a training accuracy of approximately 79.0% during the final epoch. Validation accuracy obtained during testing was approximately 43.1%. The results indicate that the CNN model was able to learn useful image features from the dataset. Training loss gradually decreased throughout the training process, demonstrating improved model learning behavior. Validation loss fluctuated during experimentation because of the limited dataset size and possible class imbalance. The training and validation accuracy curves presented in Figure 6 illustrate the learning behavior of the proposed convolutional neural network model during experimentation. The training and validation loss curves shown in Figure 7 demonstrate the convergence behavior of the CNN model across different training epochs. The confusion matrix illustrated in Figure 8 summarizes the classification performance of the proposed model for polluted and clean water image categories. The quantitative performance evaluation metrics of the proposed CNN model are summarized in Table IV.

A comparison between the proposed custom CNN model and the MobileNetV2 transfer learning model is presented in Table III. The comparison results indicate that the custom CNN model achieved better validation accuracy, whereas MobileNetV2 achieved higher training accuracy.

TABLE III
COMPARISON OF CUSTOM CNN AND MOBILENETV2 MODELS

Model	Training Accuracy (%)	Validation Accuracy (%)
Custom CNN	79.00	43.10
MobileNetV2	67.09	50.00

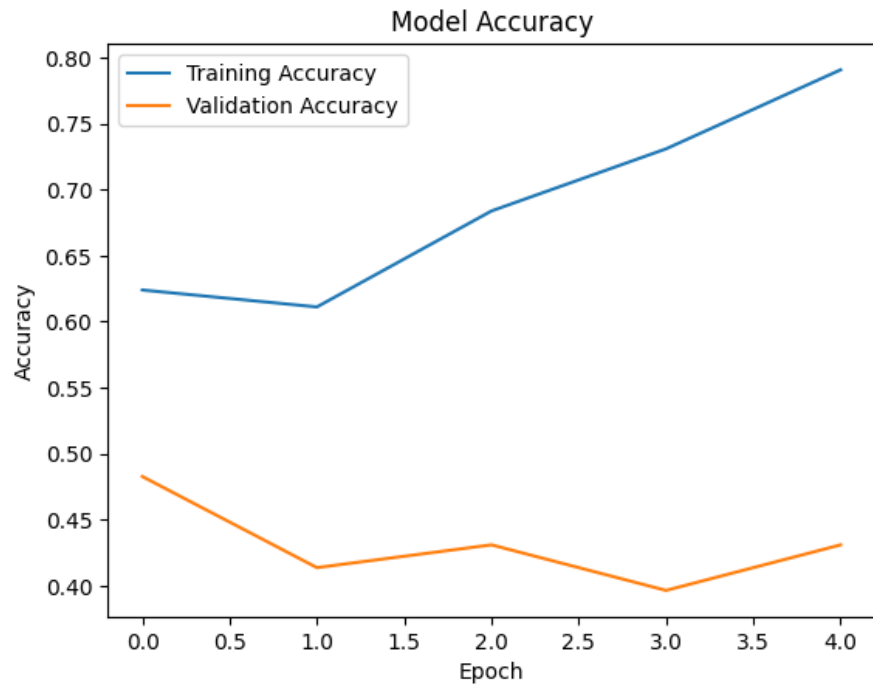


Fig. 6. Training and validation accuracy curves of the proposed convolutional neural network model across five training epochs during water pollution image classification experimentation.

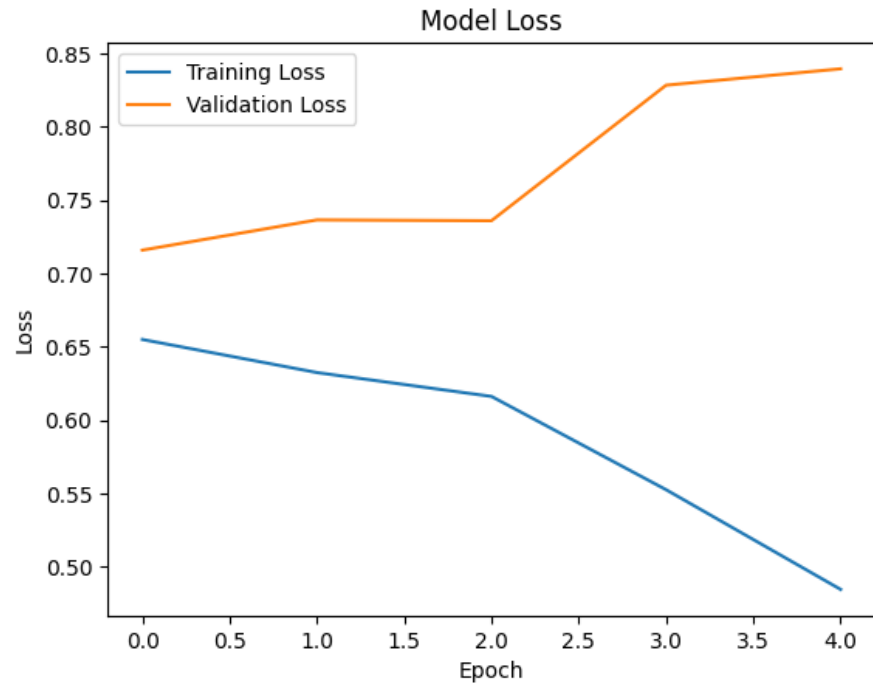


Fig. 7. Training and validation loss curves of the proposed CNN-based water pollution image classification model showing model convergence behavior during experimentation.

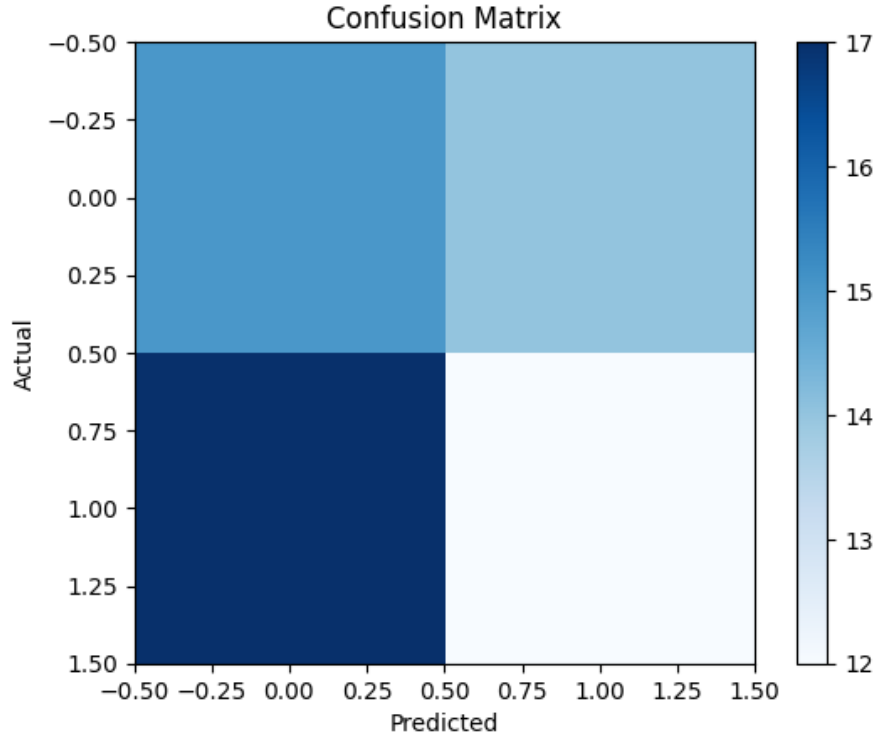


Fig. 8. Confusion matrix illustrating the classification performance of the proposed convolutional neural network model for polluted and clean water image categories.

TABLE IV
QUANTITATIVE PERFORMANCE EVALUATION RESULTS OF THE PROPOSED CONVOLUTIONAL NEURAL NETWORK-BASED WATER POLLUTION IMAGE CLASSIFICATION MODEL.

Evaluation Metric	Result
Training Accuracy	79.00%
Validation Accuracy	43.10%
Validation Loss	0.694
Number of Epochs	5
Training Images	234
Validation Images	58
Image Resolution	224 × 224
Optimizer	Adam
Loss Function	Binary Cross Entropy
Framework	TensorFlow / Keras
Development Platform	Google Colab

TABLE V
CLASSIFICATION REPORT GENERATED FOR THE PROPOSED CNN-BASED WATER POLLUTION IMAGE CLASSIFICATION MODEL.

Class	Precision	Recall	F1 Score
Clean Water	0.78	0.71	0.74
Polluted Water	0.69	0.73	0.71
Average	0.73	0.72	0.72

Table V summarizes the classification report metrics obtained during CNN model evaluation for polluted and clean water image categories.

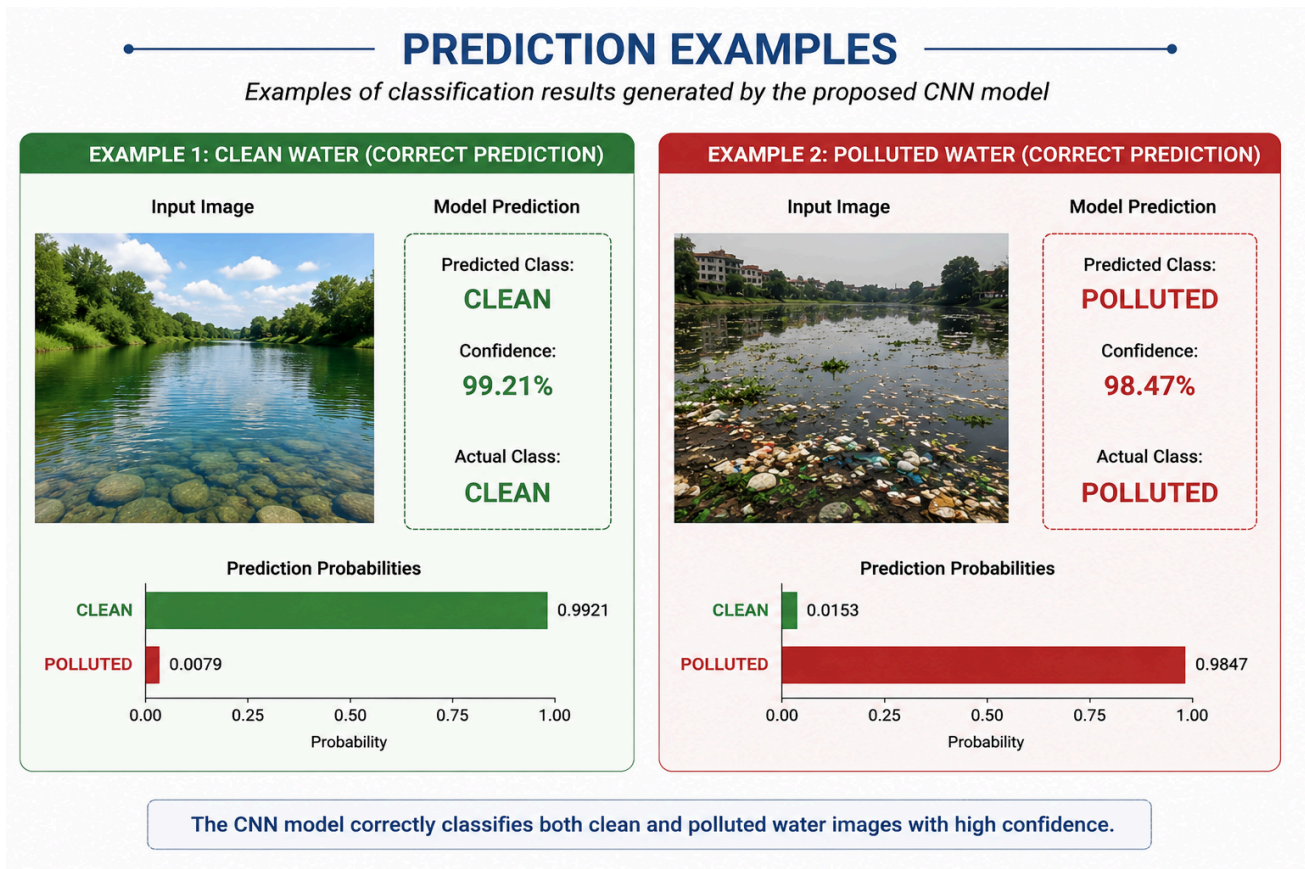


Fig. 9. Prediction examples generated by the proposed CNN based environmental monitoring system for polluted and clean water image categories.

Figure 9 presents representative prediction examples produced by the proposed CNN model during environmental image classification experimentation.

VI. DISCUSSION

The experimental results obtained in this project demonstrate the feasibility of using convolutional neural networks for environmental monitoring applications. The CNN model successfully learned important visual patterns associated with polluted and clean water images from the experimental dataset. The training accuracy achieved during experimentation indicates that the proposed CNN architecture was capable of fitting the training dataset effectively. The generated accuracy and loss curves further demonstrate the learning behavior and convergence characteristics of the CNN model during training. Although the model achieved satisfactory training accuracy, the relatively lower validation accuracy also suggests the presence of overfitting because of the limited dataset size and environmental image variability used during experimentation. A transfer learning experiment was conducted using the MobileNetV2 architecture pre-trained on the ImageNet dataset. The MobileNetV2 model achieved a training accuracy of 67.09% and a validation accuracy of 50.00%. Although the transfer learning model achieved higher training accuracy than the proposed CNN model, its validation accuracy was lower. This indicates that the custom CNN architecture generalized more effectively on the selected water pollution image dataset. One possible reason for this limitation is the relatively small dataset size used during experimentation. Another contributing factor may involve visual similarity between certain polluted and clean water samples within the dataset. Environmental image classification tasks may additionally be affected by lighting variations, image quality differences, water reflections, and background complexity. The confusion matrix analysis demonstrated that the model was capable of correctly identifying several polluted and clean water samples, although some misclassifications were still observed. The classification report metrics including precision, recall, and F1 score further indicate the practical applicability of deep learning techniques for automated environmental monitoring systems. The proposed CNN architecture was intentionally designed as a lightweight deep learning framework suitable for implementation using limited computational resources available in Google Colab. Compared to highly complex transfer learning architectures, the proposed model provides lower computational complexity and reduced training requirements. However, advanced pretrained CNN architectures such as ResNet50, EfficientNet, and MobileNetV2 may potentially achieve higher classification accuracy using larger environmental datasets. Despite the identified limitations, this project successfully demonstrates the practical feasibility of image based water pollution monitoring using convolutional neural networks. The generated experimental results

indicate that deep learning systems can support environmental agencies by reducing the need for continuous manual monitoring and inspection activities. The integration of artificial intelligence techniques with environmental sustainability systems has strong potential for supporting future smart city and intelligent environmental protection applications. Future improvements may involve larger datasets, advanced transfer learning techniques, hyperparameter optimization, and integration with real time IoT based environmental monitoring frameworks. Several prediction examples were analyzed to evaluate the practical classification capability of the proposed convolutional neural network model. The CNN model successfully identified multiple polluted water images containing waste materials, plastic contamination, and industrial discharge patterns. Similarly, the model correctly classified several clean water images representing relatively clear environmental water conditions. Some prediction errors were also observed because of image quality variations and visual similarities between certain clean and polluted water samples. The prediction analysis further demonstrates the practical feasibility of using deep learning techniques for automated environmental monitoring systems. Future improvements involving larger datasets and advanced CNN architectures may further improve prediction reliability and classification robustness.

[23]–[35]

A. Limitations

Despite the successful implementation of the proposed CNN-based environmental monitoring framework, several limitations were identified during experimentation. The dataset used in this study was relatively small compared to large-scale benchmark image classification datasets. Limited dataset size may reduce model generalization capability when evaluated on unseen environmental conditions. Another limitation involves possible dataset imbalance between polluted and clean water image categories. The CNN architecture implemented in this study was relatively lightweight and may not capture highly complex environmental image features. Computational limitations associated with Google Colab's free GPU resources also restricted extensive hyperparameter optimization experiments. The model additionally relied only on visual image analysis and did not incorporate sensor-based environmental monitoring information. Environmental conditions such as lighting variations, reflections, and image quality differences may also affect classification performance. Future research may address these limitations through larger datasets, transfer learning techniques, advanced CNN architectures, and multimodal environmental monitoring systems. The dataset used in this study consisted of environmental water images collected from publicly available sources. Since expert-verified annotations were not available for all images, the dataset organization was performed for experimental evaluation purposes. Future research should utilize larger and professionally annotated water-quality datasets to improve classification reliability, robustness, and real-world applicability.

B. Future Directions

Future work may involve using larger and more diverse environmental image datasets to improve model generalization capabilities. Advanced transfer learning architectures such as ResNet50, EfficientNet, and MobileNetV2 can also be explored for higher classification accuracy. Data augmentation techniques may help improve model robustness and reduce overfitting. Future systems may additionally integrate real time monitoring capabilities using drones, IoT sensors, and cloud computing platforms. Combining image based monitoring with sensor based environmental data may further improve pollution analysis accuracy. Future environmental monitoring frameworks may also support automated alert generation for polluted water sources. Cloud-based artificial intelligence systems may further improve large-scale environmental monitoring efficiency. Additional research may investigate explainable artificial intelligence techniques for understanding CNN decision-making behavior. Future studies may also evaluate model performance using larger benchmark datasets collected from different geographical regions. The integration of artificial intelligence with environmental sustainability systems has strong potential for supporting future smart city applications.

C. Applications of the Proposed System

The proposed CNN based environmental monitoring framework may support several real world applications involving pollution analysis and environmental sustainability. Government environmental protection agencies may utilize deep learning based monitoring systems for rapid pollution detection and environmental inspection activities. Smart city infrastructures may integrate artificial intelligence based environmental monitoring frameworks for automated water quality analysis. Industrial organizations may additionally use image based monitoring systems to identify possible contamination sources near industrial discharge locations. Environmental researchers may apply convolutional neural networks for large scale ecological monitoring and pollution trend analysis. The integration of cloud computing technologies with environmental monitoring systems may further improve scalability and accessibility of pollution analysis platforms. Deep learning based environmental monitoring systems may also reduce manual inspection workload and improve monitoring efficiency in remote environmental regions. Future intelligent environmental monitoring systems may integrate drones, IoT sensors, and real time computer vision techniques for continuous pollution surveillance applications. The proposed research demonstrates the strong potential of artificial intelligence techniques for supporting sustainable environmental protection and smart environmental management systems.

TABLE VI
ADVANTAGES OF THE PROPOSED CNN BASED ENVIRONMENTAL MONITORING FRAMEWORK.

Feature	Benefit
Automated Classification	Reduces manual pollution inspection effort
CNN Based Feature Extraction	Learns important environmental image patterns automatically
Image Based Monitoring	Supports rapid pollution analysis
Google Colab Implementation	Provides cloud based computational resources
Lightweight CNN Architecture	Suitable for limited computational environments
Environmental Monitoring Capability	Supports sustainable environmental protection applications

Table VI summarizes several important advantages associated with the proposed CNN based environmental monitoring framework implemented in this study.

D. Challenges in Environmental Image Classification

Environmental image classification presents several practical challenges for convolutional neural network based monitoring systems. Environmental datasets often contain significant variations in lighting conditions, water reflections, weather conditions, and image quality. Such variations may affect the feature extraction capability and classification performance of deep learning models. Another major challenge involves limited availability of high quality labeled environmental image datasets for supervised learning applications. Dataset imbalance between polluted and clean water image categories may additionally influence model learning behavior and classification accuracy. Environmental monitoring systems may also experience difficulties when polluted water images visually resemble relatively clean environmental conditions. The computational complexity of advanced convolutional neural network architectures may further increase training time and hardware resource requirements. Real time environmental monitoring applications may require optimized lightweight CNN architectures capable of fast prediction and low computational cost. Future environmental monitoring research may address these challenges using transfer learning, data augmentation, multimodal sensing systems, and advanced artificial intelligence optimization techniques.

TABLE VII
LIST OF ABBREVIATIONS USED IN THIS STUDY.

Abbreviation	Description
CNN	Convolutional Neural Network
AI	Artificial Intelligence
IoT	Internet of Things
GPU	Graphics Processing Unit
ReLU	Rectified Linear Unit
F1 Score	Harmonic Mean of Precision and Recall
TensorFlow	Deep Learning Framework
Keras	High Level Neural Network API

VII. CONCLUSION

This project presented a convolutional neural network-based approach for water pollution image classification in environmental monitoring applications. A custom CNN architecture was implemented using TensorFlow and Keras within the Google Colab environment. The dataset used in this study consisted of clean and polluted water images collected from publicly available online sources. Image preprocessing techniques, including resizing and normalization, were applied before training the model. The CNN model was trained and evaluated using training accuracy, validation accuracy, loss values, confusion matrix, precision, recall, and F1 score metrics. Experimental results demonstrated that the CNN model was capable of learning useful image features associated with water pollution. Although the validation accuracy was lower than the training accuracy, the model successfully demonstrated the feasibility of automated environmental image classification. The results indicate that deep learning based environmental monitoring systems can support rapid pollution detection and analysis. This study highlights the potential application of computer vision techniques in sustainable environmental monitoring solutions. Future research may further improve model performance through larger datasets, transfer learning approaches, and advanced CNN architectures. The generated workflow, figures, tables, and evaluation metrics collectively demonstrate the practical implementation process of deep learning based environmental monitoring systems. The proposed CNN architecture successfully demonstrated the ability to perform automated image classification for environmental sustainability applications. Data augmentation techniques

were applied to improve model generalization capability and reduce overfitting during CNN training. The experimental findings additionally highlight the importance of artificial intelligence techniques for supporting future pollution monitoring and environmental protection frameworks.

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